PYKV: A Scalable In-Memory Key-Value Store with Persistence

Milestone 1

- ABSTRACT

PyKV is a scalable in-memory key-value store designed for high performance and fast data access. It supports concurrent client requests using asynchronous programming. The system is built using FastAPI and provides basic operations like storing and retrieving data. This milestone focuses on setting up the core system and implementing basic functionalities.

- PROBLEM STATEMENT

Traditional systems may not handle multiple users efficiently. This project solves the problem by creating a fast and scalable key-value store using asynchronous operations.

- MODULES

Core Data Store Module

FastAPI Server Module

- WEEKLY PLAN

Week 1: Learn asyncio and setup project

Week 2: Implement GET and POST APIs

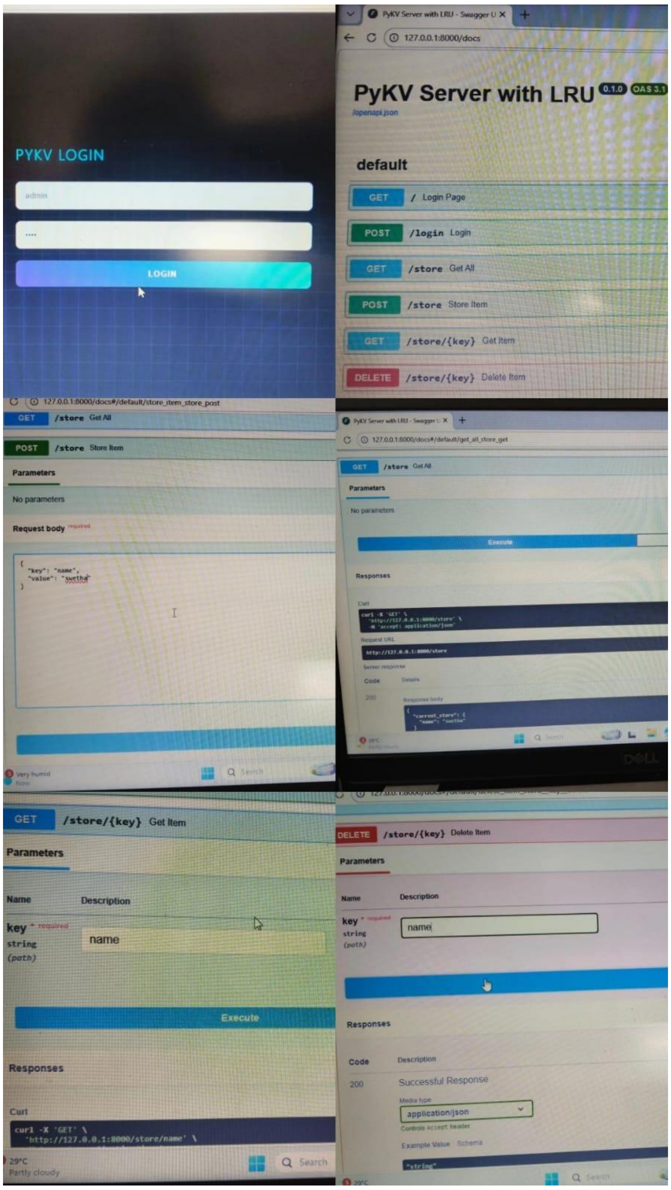
- WORK DONE

Implemented basic in-memory dictionary for storing key-value pairs. Developed FastAPI server and created API endpoints for GET and POST operations.

- PROBLEMS FACED

Faced issues with async functions and API testing. Solved by debugging and understanding FastAPI syntax.

- ♦ OUTPUT SCREENSHOT



- ♦ CONCLUSION

Successfully built a basic key-value store and tested API operations.